# A Reproduction Audit of *Deep Autoencoder-Based Anomaly Detection of Electricity Theft Cyberattacks in Smart Grids*

ATK Evidence

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**Abstract**

We implemented the paper's written ISET FC-SAE method from a clean reader's perspective and compared it with Table III. The reported balanced accuracy was 83.00%; the audited run produced 40.18%. A finite geometric calculation then showed that no change to the fitted weights, seed, or optimizer can raise this same Softmax-output method above 50.93% balanced accuracy on the fixed prepared inputs. Replacing Softmax with Sigmoid removes that particular bound, but does not reproduce the result: in a small paired fit, the best detection rate at no more than 15% false alarms was 9.74935% in the paper's score direction and 25.39063% with the direction reversed, versus 81% reported. The Sigmoid loss was still improving at epoch ten, so this is not a proof against every Sigmoid configuration. Four proposed Table III models and the mechanism, Table II, and Tables IV–V checks remain. The current conclusion is therefore narrow: one numerical result did not reproduce; one fixed-input Softmax target is excluded; and a bounded Sigmoid diagnostic did not rescue it.

## 1 The three questions

The audit keeps three questions separate.

1. **Numerical:** does the method as written reproduce the reported numbers?
2. **Mechanistic:** does the added architecture demonstrate the capability claimed to explain its advantage?
3. **Attainability:** does the reported target lie inside a justified performance envelope, or far outside it with no ordinary route to the target?

A numerical miss is not automatically a mechanism failure or an attainability result. None of the three, alone or together, identifies author intent.

## 2 What the paper says

For the ISET experiment, benign half-hourly electricity profiles are used to construct six families of synthetic attacks. The anomaly detector is trained on benign profiles. Table I describes an FC-SAE with dimensions $48 \to 400 \to 300 \to 200 \to 100 \to 100 \to 200 \to 300 \to 400 \to 48$, hidden Sigmoid activations, dropout 0.4, a Softmax output, Adam at 0.001, and mean squared reconstruction error. A reconstruction score above 0.58 is treated as malicious. Table III reports 81% detection, 15% false alarms, 85% specificity, 81% precision, 83% accuracy, 81% F1, and 81% AUC for ISET FC-SAE.

The source is incomplete or contradictory in several consequential places. For example, the paper does not make every data mapping or threshold-selection operation executable; its VAE equation mixes incompatible variables; its AEA algorithm and prose disagree; and some published metric rows cannot all obey the stated formulas. We preserve those facts and label each executable completion instead of silently choosing a favorable repair.

## 3 Clean-reader implementation

The first eligible run used 1,500,523 benign training profiles. The prepared test set contained 4,380,387 benign and 4,504,602 attack profiles after the declared ADASYN step, or 8,884,989 rows. The network had 450,448 parameters. Training stopped after 28 epochs and restored epoch 23. Fitting took 8:02:46 on one NVIDIA P100; the complete job took 9:14:27.

The implementation is intentionally literal. Corrections are separate experiments. All seven values below use the paper's printed metric definitions and the declared score direction.

Table 1: Table III ISET FC-SAE: reported and audited values (percent).

| | DR | FA | SP | PR | ACC | F1 | AUC |
|---|---|---|---|---|---|---|---|
| Reported | 81.00 | 15.00 | 85.00 | 81.00 | 83.00 | 81.00 | 81.00 |
| Audited | 25.48 | 45.13 | 54.87 | 36.73 | 40.18 | 30.09 | 39.40 |

Numerical finding. This one clean-reader FC-SAE result did not reproduce Table III. It is one eligible model and one seed, not a verdict on the complete paper.

## 4 Could a better cutoff rescue the same Softmax model?

We first asked a broader question than ordinary threshold tuning: even if we could choose the most favorable Softmax output separately for every attack, could enough attacks cross a threshold while keeping false alarms at or below 15%?

For a prepared input $x$, let $r$ be any possible 48-way Softmax output and let the reconstruction score be

$$s(x, r) = \frac{\|x - r\|_2^2}{48}.$$

Every Softmax output lies on the probability simplex. The minimum possible score is the squared distance from $x$ to that simplex divided by 48. The maximum is attained at one of the simplex vertices:

$$L(x) = \frac{d(x, \Delta)^2}{48}, \qquad U(x) = \frac{\|x\|_2^2 + 1 - 2\min_j x_j}{48}.$$

We evaluated outward-padded float64 versions of these limits over every fixed prepared row, then exhaustively enumerated the resulting score boundaries. This is conservative floating-point evidence, not certified interval arithmetic.

The reported pair is 81% detection at 15% false alarms. These limits include every set of weights, seed, optimizer, and training duration that retains the fixed prepared inputs, 48-way Softmax output, MSE score, and the paper's score direction. Ordinary training or cutoff search cannot bridge this gap inside that declared system.

This does not prove that every conceivable interpretation fails. Changing the input preparation, output activation, score, or population changes the system and requires a separately labeled experiment.

Table 2: Best possible Softmax-output performance on the fixed prepared rows.

| Quantity | Upper limit |
|---|---|
| Detection rate at FA $\leq$ 15% | 9.25% |
| Detection rate at FA $\leq$ 15.5% | 9.72% |
| Balanced accuracy at any cutoff | 50.93% |
| AUC under the paper's score direction | 45.11% |

# 5 Source-supported alternatives

Before training another model, we enumerated material readings supported by the source. Using the paper's printed Softmax output, MSE score, high-error direction, and cutoff 0.58 yields the limits above. Two additional normalization readings cap detection at that cutoff at 29.81% and 33.96%. Neither reaches 81%.

The paper also contains wording that could motivate replacing the Softmax output with independent Sigmoid outputs. This is a real architectural change, so the Softmax proof cannot be transferred to it.

# 6 Sigmoid: range first, then a paired fit

## 6.1 No-training range check

For independent Sigmoid outputs, the attainable reconstruction-score interval is different. Across all 8,884,989 prepared evaluation rows, a label-aware upper range permits as much as 85.32587% detection at at most 15% false alarms in the high-error direction, or 93.76498% after reversing the score. These are upper limits that choose favorable allowed outputs with knowledge of the labels; they are not results produced by a trained model.

At the printed cutoff 0.58, even the best allowed high-error assignment must falsely flag at least 29.66640% of benign rows. Thus Sigmoid opens a possible route at a different cutoff, but it cannot rescue the complete printed procedure at 0.58. A permitted target is not a reproduced target.

## 6.2 Small matched training control

We next ran a deliberately small control. Softmax and Sigmoid used the same architecture, 450,448 parameters, hidden activations, dropout, optimizer, 2,048 fitting rows, 1,024 benign calibration rows, ten epochs, 640 updates, and identical starting weight arrays. Only the final activation differed. The held-out evaluation contained 12,119 rows: 6,144 attacks, 1,024 source benign days, and 4,951 synthetic benign rows.

Table 3: Exhaustive held-out cutoff enumeration after the paired fit (percent).

| Model and score direction | DR at 15 | DR at 15.5 | Best ACC | AUC |
|---|---|---|---|---|
| Softmax, high error | 8.64258 | 8.83789 | 50.05667 | 37.68698 |
| Sigmoid, high error | 9.74935 | 9.99349 | 50.33353 | 37.70499 |
| Softmax, low error | 25.52083 | 26.12305 | 61.48292 | 62.31302 |
| Sigmoid, low error | 25.39063 | 25.81380 | 61.64614 | 62.29501 |

For the fitted Sigmoid model, all 12,060 ROC boundaries were enumerated. In the paper's high-error direction and at FA $\leq$ 15%, at most 896 of 5,975 benign rows can be false alarms and only 599 of 6,144 attacks are detected. No threshold rounding can turn 9.74935% into 81%. At cutoff 0.58 the fitted model gives 25.24414% detection and 47.02929% false alarms. A threshold set using only the benign calibration rows gives 7.12891% detection and 9.92469% false alarms.

Sigmoid therefore did not rescue this bounded fit. It is not an all-Sigmoid proof. Its calibration MSE improved from 1.61028 at epoch one to 1.33863 at epoch ten, and epoch ten was the selected checkpoint. No long-run plateau has been established.

## 7 What useful work did the added output perform?

The paired control isolates one proposed change. In the paper's score direction, Sigmoid improves constrained detection by about 1.11 percentage points over Softmax, from 8.64258% to 9.74935%. The reversed direction is slightly worse. This bounded result shows no material capability gain approaching the reported target, but it does not yet test the paper's main FC-versus-LSTM mechanism.

A mechanism claim needs a task that distinguishes the proposed capability. The next matched tests therefore keep parameter budget and training protocol as close as possible while comparing FC and recurrent models, then destroy or permute temporal order. If an LSTM advantage is attributed to temporal structure, that advantage should respond predictably when temporal structure is removed.

## 8 Static checks on the broader paper

Independent of training, five Table II rows and eight Table III rows are inconsistent with a common balanced-evaluation prevalence under the paper's printed DR, FA, and precision definitions, even after allowing for rounding. Table II's Naive Bayes row reports DR = PR = 75% but F1 = 77%; their harmonic mean is 75%. Table V reports different false-alarm rates for each attack type within a model. If the model, threshold, and benign evaluation population are fixed, false alarms cannot depend on which attack family is paired with those same benign rows. These are source inconsistencies. They do not by themselves show how the experiments were implemented.

## 9 Current coverage and next work

| Paper result | Current eligible coverage | Next evidential question |
|---|---|---|
| Table III FC-SAE | One full clean-reader run; Softmax fixed-input bound; paired Sigmoid diagnostic | Numerical miss and conditional unattainability are established at their stated scope. |
| Table III LSTM-SAE, FC-VAE, LSTM-VAE, LSTM-AEA | Not yet run through the current clean-reader path | Do the other four proposed models reproduce their rows? |
| FC versus LSTM mechanism | Not yet tested in a matched mechanism experiment | Does recurrence add useful temporal capability, and does any advantage survive temporal destruction? |
| Tables IV–V | Historical exploratory artifacts exist, but current eligible coverage is incomplete | For models that pass the frozen eligibility gate, do timing and per-attack results reproduce? |

| Paper result | Current eligible coverage | Next evidential question |
|---|---|---|
| Table II | Literal route is not executable because roughly 1,034 daily values must become 48 inputs without a stated mapping | Do any predeclared, materially reasonable source-supported mappings reproduce the table? |

The order is breadth first: complete one declared run for each remaining proposed family before deep seed or hyperparameter searches; then run the matched mechanism tests; then Tables IV–V for eligible models; then the finite Table II interpretation family. Every expensive run must answer a named uncertainty under a frozen data, code, seed, metric, budget, and stopping rule.

## 10 Conclusion

The present evidence supports three carefully bounded statements. First, the clean-reader ISET FC-SAE implementation did not reproduce its Table III row. Second, for the fixed prepared inputs, a 48-way Softmax output scored by MSE cannot attain the reported operating point, regardless of training. Third, a small matched Sigmoid fit remained far below the target, although its improving loss means broader Sigmoid attainability remains open.

The stronger paper-level conclusion is not yet available. The remaining model families and the mechanism-specific tests are necessary to decide whether the same pattern extends across the paper. Undocumented code or an untested method outside the declared interpretation space also remains possible. No current result warrants a claim about fabrication or author intent.

## References


1. B. Takiddin, M. Ismail, U. Zafar, and E. Serpedin, "Deep Autoencoder-Based Anomaly Detection of Electricity Theft Cyberattacks in Smart Grids," *IEEE Systems Journal*, 2022.

2. Commission for Energy Regulation, Smart Metering Project electricity customer behaviour trial data, accessed through the Irish Social Science Data Archive under its terms.

3. ATK Evidence, frozen clean-reader implementation, experiment contracts, result records, and source-to-code audit in the accompanying repository.